Lucas A. Saavedra

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EDUCATION

Software Engineering, Pontifical Catholic University of Argentina (UCA), Buenos Aires, Argentina, March 2018-December 2023

WORK EXPERIENCE

Back-End Developer, Deloitte, Buenos Aires, Argentina, January 2021-June 2022

- *Threat Intelligence Back-end Developer.*
- *I worked developing Codex Gigas, a malware "DNA" profiling search engine discovers malware patterns and characteristics assisting individuals who are attracted in malware hunting, and ImageSimilarity, a tool that divides a group of images according to their similarity grade using the SSIM method. Also, I developed ThreatConnect apps.*
- *Python, Docker, MongoDB, and SQL were used for development.*

Research Assistant, Molecular Neurobiology Laboratory, BIOMED UCA-CONICET, (Prof. Francisco J. Barrantes, head), Pontifical Catholic University of Argentina (UCA), Buenos Aires, Argentina, January 2022-to date

- *Implementation of Deep Learning models for the analysis of static and dynamic properties of the receptor for the neurotransmitter acetylcholine (nAChR) with orientation to graph theory, artificial neural networks and geometric computing.*
- *Analysis of distribution and density of nicotinic acetylcholine receptors (nAChRs) in the plasma membrane using super-resolution microscopy (STED and STORM). Advanced analytical methods implemented with MATLAB, based on graph theory and computational geometry to study the topography of nanoclusters.*
- *Development of efficient and autonomous algorithm based on binary classifications with supervised Graph Neural Networks (GNNs) to detect and quantify the dynamic clustering of particles in membrane proteins in single-molecule microscopy (SMLM) data. The algorithm was developed with TensorFlow.*

TEACHING EXPERIENCE

Assistant professor: Data oriented programming, Pontifical Catholic University of Argentina (UCA), Buenos Aires, Argentina, July 2024-to date.

- *Object Oriented Programming (OOP).*
- *Technologies: NumPy , Pandas, Python, Java, SQL.*
- *Software design with UML.*

Assistant professor: Discrete mathematics, Pontifical Catholic University of Argentina (UCA), Buenos Aires, Argentina, March 2024-to date.

- *Number theory.*
- *Graph theory.*
- *Demonstration methods.*
- *Boolean algebra.*

Assistant professor: Introduction to Data Science, Pontifical Catholic University of Argentina (UCA), Buenos Aires, Argentina, March 2023-to date.

- *Definitions and concepts of Artificial Intelligence.*
- *Introduction to artificial neural networks.*
- *Scientific method.*

PREPRINTS

Reina, F., Saavedra, L. A., Eggeling, C., & Barrantes, F. J. (2025). Concurrent diffusion of nicotinic acetylcholine receptors and fluorescent cholesterol disclosed by two-colour sub-millisecond MINFLUX-based single-molecule tracking [Preprint]. Research Square (Nature portfolio). https://www.researchsquare.com/article/rs-5619606/v1

FULL-LENGTH PUBLICATIONS IN INTERNATIONAL PEER REVIEWED JOURNALS

Saavedra, L. A., Buena-Maizón, H., & Barrantes, F. J. (2022). Mapping the Nicotinic Acetylcholine Receptor Nanocluster Topography at the Cell Membrane with STED and STORM Nanoscopies. International Journal of Molecular Sciences, 23(18), 22. https://doi.org/10.3390/ijms231810435

Saavedra, L. A., Mosqueira, A., & Barrantes, F. J. (2024). A supervised graph-based deep learning algorithm to detect and quantify clustered particles. Nanoscale, 16(32), 15308-15318. https://doi.org/10.1039/D4NR01944J

ABSTRACTS

Barrantes, F. J., Mosqueira, A., Buena-Maizón, H., & Saavedra, L. A. (2023). The individual molecule and its tribe, the nanocluster: the case of the nicotinic acetylcholine receptor. 11th International Weber Symposium. Punta del Este, Uruguay.

Saavedra, L. A. & Barrantes, F. J. (2024). Machine learning empowers static and dynamics clustering characterization of transmembrane receptors. Building Bridges in Computational Biophysics (BBCB) v3.0.